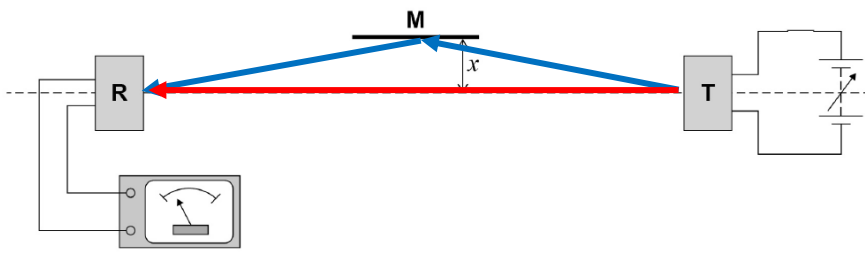
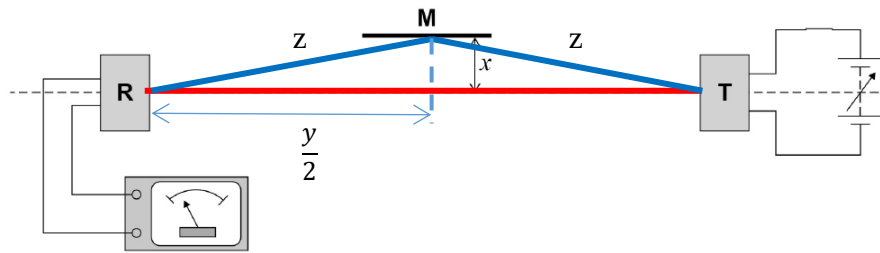


8	(a)		π radians	1
	(b)	(i)	 Award 1 mark for each correct path.	2
	(b)	(ii)	Resultant displacement/amplitude is the vector sum of the resultant displacement/amplitude of the waves. (It is “considering”, hence, do not need to be too strict on the words) The two waves reaching the receiver have different amplitude/energy/power/intensity. Because (any one of the possible reasons) 1. the distance travelled by the two waves are different. 2. some energy is absorbed upon reflection. Hence no complete cancellation / resultant displacement/amplitude is not zero.	1 1 1
	(c)	(i)	Measure the distance of the two ends of plate M And adjust (make sure) plate M such that the two ends are equidistant from the marked line. OR Use a set square to align a ruler to be perpendicular to the reference line. Move the set square along the ruler to align plate M at the new position.	1 1

(ii)

Recognising the path difference

Path difference = $2z - y$

1 mark to recognise the path difference

1

$$= 2\sqrt{x^2 + \left(\frac{y}{2}\right)^2} - y$$

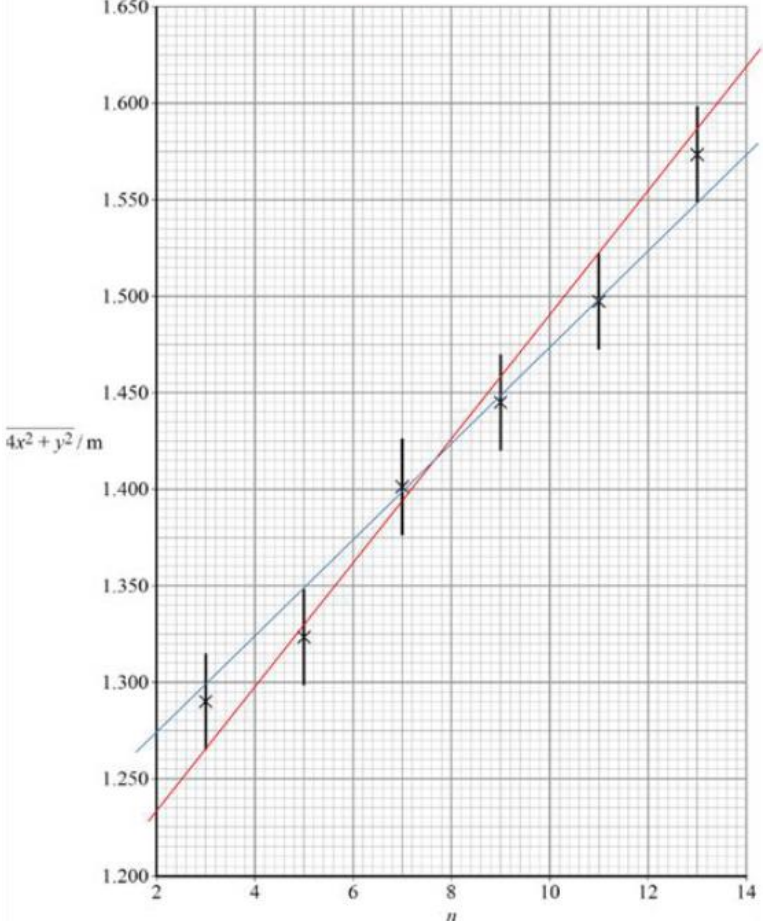
1 mark to be able to write the equation in terms of the hypotenuse of the triangle

1

$$= \sqrt{4x^2 + (y)^2} - y$$

1

Since there is a phase change at the surface, for destructive interferencePath difference = $n\lambda$ 1 mark to explain why path difference is equated to $n\lambda$

	(d)	(i)	 $G_{\max}$ line is drawn through bottom of $n=3$ error bar and through top of $n=11$ error bar. Gradient $G_{\max} = 0.0312$ to 0.0327	1 1
		(ii)	$G_{\min}$ line is drawn through top of $n=5$ error bar and through bottom of $n=13$ error bar. Gradient $G_{\min} = 0.0242$ to 0.0257	1 1
		(iii)	average $\lambda = \frac{G_{\max} + G_{\min}}{2}$ Allow ECF	1
		(iv)	It is in band C. (allow ecf)	1

		(v) $\Delta\lambda = G_{\max} - \lambda_{\text{ave}} = 0.36 \times 10^{-2}$ Percentage uncertainty in $\lambda = \frac{\Delta\lambda}{\lambda} \times 100$ $= 12.6 \%$ Allow ECF	1 1
	(e)	$y = kx$ $4.5 = k_1 (2.9) \quad 22.3 = k_2 (5.8) \quad 55.5 = k_3 (9.0) \quad 99.5 = k_4 (12.0)$ $k_1 = 1.5 \quad k_2 = 3.8 \quad k_3 = 6.1 \quad k_4 = 8.29$ Conclusion: Since the k values are significantly different, y is not proportional to x. (1 mark for showing at least two k values with correct calculation 1 mark for conclusion)	1 1
	(f)	Because the wave reflecting from plate M will not reach the receiver. Hence, there will not be any superposition of the waves at the receiver. Or if student mentioned about angle of incidence = angle of reflection for the second mark.	1 1